

GCD Recursion Theorem

1 Why We Need a Better Method

Goal

We want to compute

$$\gcd(a, b)$$

efficiently for nonnegative integers a and b .

Because

$$\gcd(a, b) = \gcd(|a|, |b|),$$

it is enough to study nonnegative inputs.

Factoring is not efficient

In principle, we can compute $\gcd(a, b)$ from the prime factorizations of a and b . But factoring large integers is not known to be polynomial-time, so this is not a good algorithmic strategy.

2 GCD from Prime Factorization

Common Prime Basis

Suppose a and b use the same prime list

$$p_1, p_2, \dots, p_r,$$

with zero exponents allowed when a prime does not appear in one number.

Write

$$a = p_1^{e_1} p_2^{e_2} \cdots p_r^{e_r},$$

and

$$b = p_1^{f_1} p_2^{f_2} \cdots p_r^{f_r}.$$

GCD Factorization Formula

$$\gcd(a, b) = p_1^{\min(e_1, f_1)} p_2^{\min(e_2, f_2)} \cdots p_r^{\min(e_r, f_r)}.$$

The reason is simple: a common divisor can use only the prime powers that appear in both a and b . For each prime p_i , the largest exponent available to both numbers is

$$\min(e_i, f_i).$$

Example

Let

$$a = 2^4 \cdot 3^2 \cdot 5, \quad b = 2^2 \cdot 3^5 \cdot 7.$$

Using zero exponents for missing primes:

$$a = 2^4 \cdot 3^2 \cdot 5^1 \cdot 7^0,$$

$$b = 2^2 \cdot 3^5 \cdot 5^0 \cdot 7^1.$$

Therefore,

$$\gcd(a, b) = 2^{\min(4,2)} 3^{\min(2,5)} 5^{\min(1,0)} 7^{\min(0,1)}.$$

So

$$\gcd(a, b) = 2^2 \cdot 3^2 = 36.$$

3 GCD Recursion Theorem

Theorem 31.9

For every nonnegative integer a and every positive integer b ,

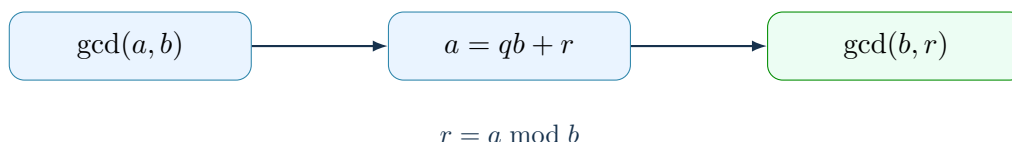
$$\gcd(a, b) = \gcd(b, a \bmod b).$$

Why this matters

This theorem is the engine behind Euclid's algorithm. It replaces a larger pair (a, b) with a smaller pair

$$(b, a \bmod b)$$

without changing the gcd.



4 Proof of the Recursion Theorem

Setup

Let

$$r = a \bmod b.$$

By the division theorem,

$$a = qb + r, \quad 0 \leq r < b.$$

Equivalently,

$$r = a - qb.$$

Proof strategy

It is enough to show that the common divisors of a, b are exactly the same as the common divisors of b, r . Then their greatest common divisor must be the same.

Direction 1: From (a, b) to (b, r)

Assume d is a common divisor of a and b :

$$d \mid a \quad \text{and} \quad d \mid b.$$

Because $r = a - qb$, we substitute the known divisibility:

$$d \mid a \quad \text{and} \quad d \mid qb.$$

Therefore d divides the difference:

$$d \mid (a - qb).$$

But $a - qb = r$, so

$$d \mid r.$$

We already have $d \mid b$, hence

$$\boxed{d \text{ divides both } b \text{ and } r.}$$

Result of Direction 1

Every common divisor of a and b is also a common divisor of b and r .

Direction 2: From (b, r) to (a, b)

Now assume d is a common divisor of b and r :

$$d \mid b \quad \text{and} \quad d \mid r.$$

Because $a = qb + r$, we use the known divisibility:

$$d \mid qb \quad \text{and} \quad d \mid r.$$

Therefore d divides the sum:

$$d \mid (qb + r).$$

But $qb + r = a$, so

$$d \mid a.$$

We already have $d \mid b$, hence

$$\boxed{d \text{ divides both } a \text{ and } b.}$$

Result of Direction 2

Every common divisor of b and r is also a common divisor of a and b .

Conclusion

The two pairs

$$(a, b) \quad \text{and} \quad (b, r)$$

have exactly the same common divisors. Therefore their largest positive common divisor is the same:

$$\boxed{\gcd(a, b) = \gcd(b, r) = \gcd(b, a \bmod b).}$$

5 Worked Trace

Problem

Use the recursion theorem to compute $\gcd(252, 105)$.

Solution

$$252 = 2 \cdot 105 + 42,$$

so

$$\gcd(252, 105) = \gcd(105, 42).$$

Next,

$$105 = 2 \cdot 42 + 21,$$

so

$$\gcd(105, 42) = \gcd(42, 21).$$

Finally,

$$42 = 2 \cdot 21 + 0,$$

so

$$\gcd(42, 21) = 21.$$

Answer

$$\gcd(252, 105) = 21.$$

6 Final Summary

Keep These

$$\begin{aligned} a &= p_1^{e_1} \cdots p_r^{e_r}, \quad b = p_1^{f_1} \cdots p_r^{f_r} \\ \gcd(a, b) &= p_1^{\min(e_1, f_1)} \cdots p_r^{\min(e_r, f_r)} \\ a &= qb + r, \quad 0 \leq r < b, \quad r = a \bmod b \\ \gcd(a, b) &= \gcd(b, a \bmod b) \end{aligned}$$